Network Monitoring Lab

Cisco Packet Tracer Project Documentation

Syslog • CDP • LLDP • SSH • IPv4 Routing • Network Monitoring

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Project Type: CCNA / Network Engineer / NOC

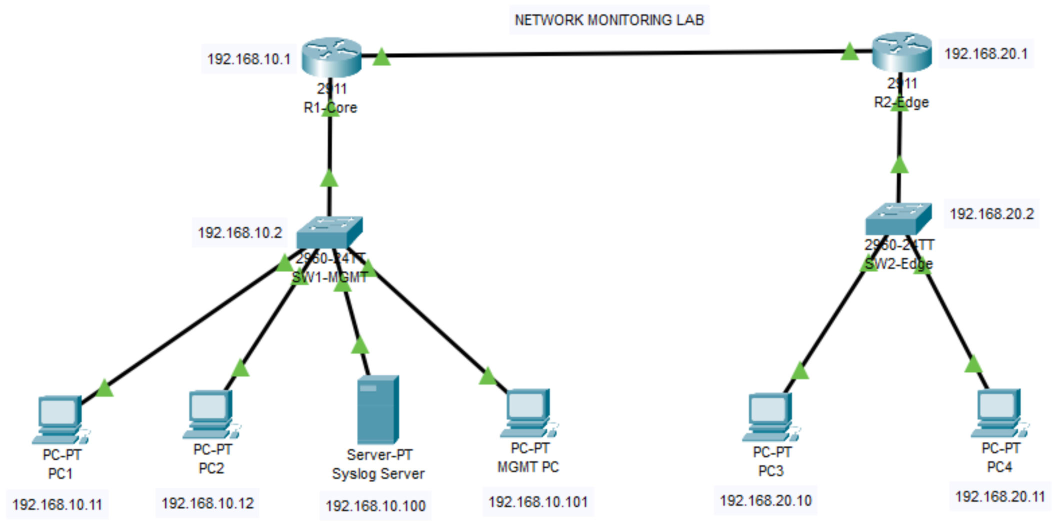
Portfolio Project Platform: Cisco Packet Tracer

1. Project Objectives

This project demonstrates a small enterprise-style network in Cisco Packet Tracer with centralized Syslog logging, Cisco Discovery Protocol (CDP), Link Layer Discovery Protocol (LLDP), secure remote administration using SSH, IPv4 addressing, static routing, and verification of network events.

The lab is designed to simulate practical network monitoring and management tasks performed by Network Engineers and NOC Engineers. Two routers connect two LAN segments. Each LAN contains a Layer-2 switch and end devices. A dedicated server provides centralized Syslog collection, while a management PC is used to test connectivity and SSH access.

2. Network Topology



Physical Connections

R1 G0/0 → SW1 G0/1

R1 G0/1 → R2 G0/1

R2 G0/0 → SW2 G0/1

SW1 Fa0/1 → PC1; Fa0/2 → PC2; Fa0/3 → Syslog Server; Fa0/4 → Management PC

SW2 Fa0/1 → PC3; Fa0/2 → PC4

3. IP Addressing Plan

Devices	Interface	IP address	Mask	Gateway
R1-Core	G0/0	192.168.10.1	255.255.255.0	-
R1-Core	G0/1	10.0.0.1	255.255.255.252	-
R2-Edge	G0/0	192.168.20.1	255.255.255.0	-
R2-Edge	G0/1	10.0.0.2	255.255.255.252	-
SW1-Mgmt	VLAN1	192.168.10.2	255.255.255.0	192.168.10.1
SW2-Edge	VLAN1	192.168.20.2	255.255.255.0	192.168.10.1

Devices	Interface	IP address	Mask	Gateway
R1-Core	NIC	192.168.10.1	255.255.255.0	-
R1-Core	NIC	10.0.0.1	255.255.255.252	-
R2-Edge	NIC	192.168.20.1	255.255.255.0	-
R2-Edge	NIC	10.0.0.2	255.255.255.252	-
SW1-Mgmt	NIC	192.168.10.2	255.255.255.0	192.168.10.1
SW2-Edge	NIC	192.168.20.2	255.255.255.0	192.168.10.1

4. Router Configuration

Router1 — R1-CORE

enable
configure terminal hostname R1-CORE
interface gigabitEthernet 0/0
ip address 192.168.10.1 255.255.255.0 no shutdown
exit
interface gigabitEthernet 0/1
ip address 10.0.0.1 255.255.255.252 no shutdown
exit
ip route 192.168.20.0 255.255.255.0 10.0.0.2

Router2 — R2-EDGE

enable
configure terminal hostname R2-EDGE
interface gigabitEthernet 0/0
ip address 192.168.20.1 255.255.255.0 no shutdown
exit
interface gigabitEthernet 0/1
ip address 10.0.0.2 255.255.255.252 no shutdown
exit
ip route 192.168.10.0 255.255.255.0 10.0.0.1

Router Verification

show ip interface brief show ip route
ping 10.0.0.2
ping 192.168.20.1

5. Switch Management Configuration

SW1-MGMT

```
enable
configure terminal hostname
SW1-MGMT
interface vlan 1
ip address 192.168.10.2 255.255.255.0
no shutdown exit
ip default-gateway 192.168.10.1 end
copy running-config startup-config
```

SW2-EDGE

```
enable
configure terminal hostname SW2-EDGE
interface vlan 1
ip address 192.168.20.2 255.255.255.0 no shutdown
exit
ip default-gateway 192.168.20.1 end
copy running-config startup-config
```

6. Centralized Syslog Configuration

On the Packet Tracer Server, open Services → SYSLOG and turn the Syslog service ON. Configure every Cisco device to send messages to 192.168.10.100.

```
logging 192.168.10.100
logging trap informational
logging buffered 16384
```

Verification

```
show logging
```

The Syslog server should display messages generated by the routers and switches. Useful events to demonstrate include interface state changes and configuration changes.

7. Cisco Discovery Protocol (CDP)

CDP is used to discover directly connected Cisco devices and display neighbor information.

```
enable
configure terminal cdp run
end
show cdp neighbors
show cdp neighbors detail
```

8. Link Layer Discovery Protocol (LLDP)

LLDP provides vendor-neutral neighbor discovery. Enable it on all Cisco devices.

```
enable
configure terminal lldp run
end
show lldp neighbors
show lldp neighbors detail
```

9. Secure SSH Remote Management

SSH is configured so that the management PC can securely access the routers and switches using the local user database.

```
enable
configure terminal
ip domain-name networklab.local
username admin privilege 15 secret Cisco@123
crypto key generate rsa 1024
ip ssh version 2
line vty 0 4 login local transport input ssh
exit
end
copy running-config startup-config
```

SSH Test

From Management PC:

```
ping 192.168.10.1
ssh -l admin 192.168.10.1
ping 192.168.10.2
ssh -l admin 192.168.10.2
```

A successful test should display the device prompt, such as R1-CORE# or SW1-MGMT#.

10. Syslog Event Generation

Interface Down / Up Event

```
configure terminal
interface gigabitEthernet 0/0 shutdown
end
! Wait for the Syslog event.
configure terminal
interface gigabitEthernet 0/0 no shutdown
end
```

Capture the resulting interface-down and interface-up messages on the Syslog Server.

Configuration Change Event

```
configure terminal
interface gigabitEthernet 0/0 description LAN-to-SW1
end
```

Verify the resulting configuration-change message on the local log and, where supported by the Packet Tracer device/service, on the centralized Syslog server.

11. End To End Verification Checklist

- All router and switch interfaces required by the topology are up/up.
- R1 can ping R2's transit address 10.0.0.2.
- R2 can ping R1's transit address 10.0.0.1.
- Management PC can reach the Syslog Server.
- Management PC can reach Router2 and PC3/PC4 across the routed network.
- show cdp neighbors displays directly connected Cisco neighbors.
- show lldp neighbors displays LLDP neighbors.